A Digital Block Control for A Photoplethysmography System in CMOS 65 nm

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***Abstract*—This paper proposes a digital block to control an LED driver, a programmable gain amplifier (PGA) and an IDAC for an analog photoplethysmography (PPG) analog front-end circuit. The paper shows the results of testbenches for the digital circuit as well as the logical and physical synthesis that can be verified for the LED driver circuit with an estimated power consumption of 73.211 nW and an estimated area of $291\ \mu m^2$. The IDAC PGA digital circuit has an estimated power consumption of 122.32 nW and an estimated area of $559.840\ \mu m^2$. For both circuits, this results in an estimated total power consumption of 195.531 nW and an estimated total silicon area of $850.84\ \mu m^2$.**

***Keywords*—Digital block, state machine, logical, physical, synthesis.**

I. Introduction

Photoplethysmography (PPG) is a vital technique used for non-invasive monitoring of various physiological parameters, such as heart rate (HR), respiratory rate (RR) and oxygen saturation [1]. Traditional PPG sensors, especially the ones that use LEDs, consume significant power, necessitating ultra-low power designs to enhance battery life without compromising performance. Those sensors have innumerable analog systems that require precision for better performance. The analog systems of those kinds of sensors can be controlled using digital blocks.

Digital blocks are programmed and not made using traditional schematic software. They are responsible for controlling some parts of the circuit and can be state machines that must control switches for the circuit to work properly and obtain the best performance for the analog blocks.

The digital block diagram used in this project is shown in Figure 1, where the digital block can control multiple blocks. This work focuses on two particular digital block systems where the controls are only the LED Driver, the IDAC and the PGA for the PPG sensor in 65nm technology.

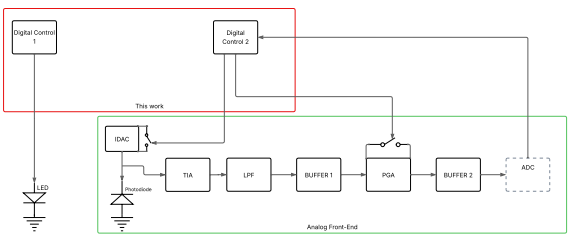

Fig. 1: Block diagram for the DBE system.

This paper is separated into Sections, where Section 2 is the digital block for the LED Driver, Section 3 is the digital block for the IDAC and PGA control, Section 4 shows the discussion and the results, and the paper is concluded in Section 4.

II. Digital Block for the LED Driver

The digital block of the LED driver is the block that will consume more power since it needs to use high currents to power the LED. The control of the LED to optimize the circuit is known for any circuit that uses LED [2]. This digital block consists of one-state machines programmed in the System Verilog language. Its main objective is to control the input current to the LED using five bits to control five switches. In this paper, we focus only on the digital block, and no LEDs were used to measure the total power consumption of the digital block. This block is shown as Digital Block 1 in Figure 1.

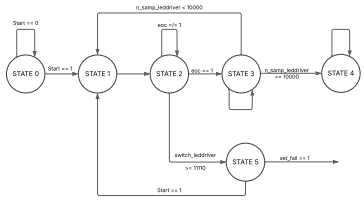

Fig. 2: State Machine Diagram for the LED Driver

The LED Driver states are:

- State 0 - Start or Loop to state 0: In this state, the state machine will remain in its state until Start goes to 1. When Start is at 1 the state machine will go to State 1.
- State 1 - State for state change: In this state, the state machine will only change state to check if the ADC has ended its conversion correctly.
- State 2 - Waiting for the result of the ADC conversion: In this state, the state machine will wait for the ADC conversion; if the ADC has not finished its conversion yet, it will stay in state 2. When the ADC finishes its conversion, it will increment the number of samples by one and check if the ADC is saturated. If the ADC is saturated, it will increment the number of switches by one. After the switch is closed, it will go to state 3. If all the switches are closed and the ADC is still saturated, it will go to state 5.

- State 3 - Waiting for the number of samples for a new sampling: In this state, the state machine will wait for sixteen samples to make a new comparison. If these sixteen samples occur, the state machine will reset the sample counter and make a comparison with the number of keys and the ADC saturation. If it is greater, it will go to state 4; if it is less, it will return to state 1.
- State 4 - OK State: In this state, the switching occurred correctly and the ADC was not saturated, so it is possible to send a flag indicating that the saturation was calibrated.
- State 5 - Failure State: In this state, the state machine attempts to control the switching, making all switching combinations, and if the ADC is still saturated, a failure occurs. Therefore, in this state, a failure flag is raised and set to 1. This flag can be used to initialize another state machine.

The results of the logical and physical synthesis of the LED Driver are shown in Section 4.

III. Digital Block for the IDAC-PGA

The IDAC-PGA digital block is a digital block programmed in System Verilog, and is characterized by being a cascaded state machine, or a series-connected state machine, which means that it is a block that contains both state machines together. This block is shown as Digital Block 2 in Figure 1.

The first state machine is the IDAC state machine; the IDAC will control the current closing five switches using five bits to do the control. Those switches will close and open when the ADC is saturated. If the IDAC state machine fails, a failure flag is raised that triggers the PGA state machine, and in this part, the state machine will control the PGA gain by doing the combination of five switches to control the resistance of the PGA. This digital block aims to control switches to control the IDAC current sources and control switches for the total resistance for the PGA gain.

This state machine consists of eleven states, which are presented below. Using this topology of digital back-end allows the circuit to use one less ADC while maintaining the control for both blocks [1] [3]. Figure 3 illustrates the state diagram of the IDAC-PGA digital block state machine.

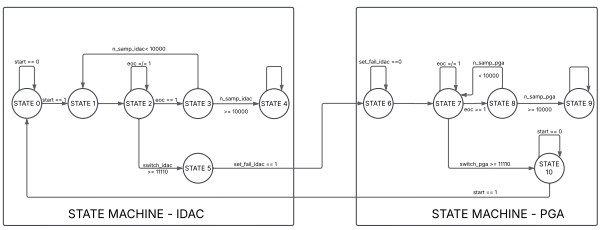

Fig. 3: State Machine Diagram - IDAC_PGA

The states for the IDAC-PGA are:

- State 0 - Start or loop to State 0: In this state, the state machine will remain in its state until Start goes to 1. When Start is at 1, the state machine will go to state 1.
- State 1 - State for state change: In this state, the state machine will only change state to check if the ADC has ended its conversion correctly.
- State 2 - Waiting for the result of the ADC to IDAC conversion: In this state, the state machine will wait for the ADC conversion; if the ADC has not finished its conversion yet, it will stay in state 2. When the ADC finishes its conversion, it will increment the number of samples by one and check if the ADC is saturated. If the ADC is saturated, it will increment the number of switches by one. After the switch is closed, it will go to state 3. If all the switches are closed and the ADC is still saturated, it will go to state 5.
- State 3 - Waiting for the number of samples for a new sampling for IDAC: In this state, the state machine will wait for sixteen samples and, if these samples are saturated, the state machine will return to state 1; otherwise, it will go to state 4.
- State 4 - OK state for IDAC: In this state, the switching occurred correctly and the ADC was not saturated, so it is possible to send a flag indicating that the saturation was calibrated. Note that if the signal is calibrated in this state, the PGA state machine will not be activated.
- State 5 - Failure state for IDAC: In this state, the state machine attempted to control the switching, making all switching combinations, and if the ADC is still saturated, a failure occurs. Therefore, in this state, a failure flag is raised. This flag is then used to start the PGA state machine.
- State 6 - Initialization state for the PGA state machine: In this state, the state machine for the PGA will be initialized.
- State 7 - Waiting for the result of the ADC to PGA conversion: In this state, the state machine will wait for the ADC conversion; if the ADC has not finished its conversion yet, it will stay in state 7. When the ADC finishes its conversion, it will increment the number of samples by one and check if the ADC is saturated. If the ADC is saturated, it will increment the number of switches by one. After the switch is closed, it will go to state 8. If all the switches are closed and the ADC is still saturated, it will go to state 10.
- State 8 - Waiting for the number of samples for a new sampling for the PGA: In this state, the state machine will wait for sixteen samples and, if these samples are saturated, the state machine will return to state 6; otherwise, it will go to state 9.
- State 9 - OK state for the PGA: In this state, the switching occurred correctly and the ADC was not saturated, so it is possible to send a flag that the saturation was calibrated. Note that this state will only occur if the IDAC state machine is unable to calibrate the signal. After reaching this state, the state machine will remain in this state.
- State 10 - PGA failure state: In this state, the state machine attempted to control the switching, making all

switching combinations, and if the ADC is still saturated, the failure occurs. Therefore, in this state, a failure flag is raised.

The results for the digital blocks are shown in the next Section.

IV. Digital Block Control Results and Layout

In this Section, we will discuss the results of the testbench made using a clock of 5MHz and the results of the logical and physical synthesis. For the digital state machine of the LED Driver, the testbench is shown in Figures 4 and 5.

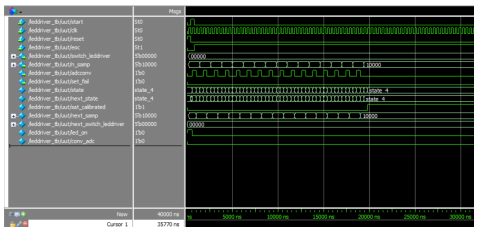

Fig. 4: Testbench for the state 4 for the LED Driver.

As shown in Figure 4, the LED Driver state machine will go to state 4 if the ADC is not saturated. From Figure 4, it's possible to see the switches closing and opening until they reach the number of sixteen.

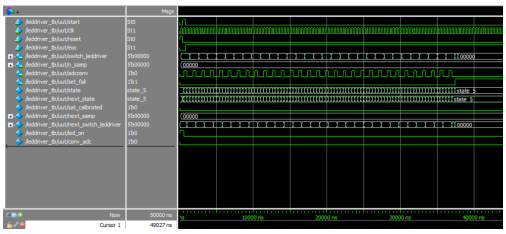

Fig. 5: Testbench for the state 5 for the LED Driver.

As shown in Figure 5, the LED Driver state machine will go to state 5 if the ADC is saturated and after making all the combinations of switches it will fail, raising the failure flag and maintaining its state in state 5 until the command to start the calibration again.

For the digital block of LED Driver, the logical synthesis is shown in Figure 6.

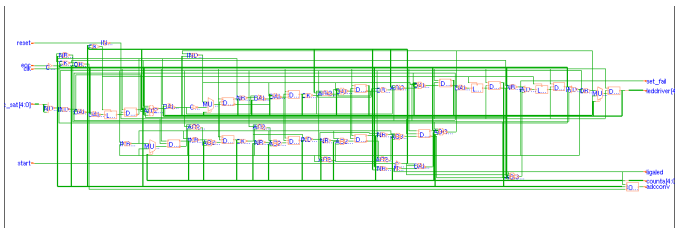

Fig. 6: Logic synthesis for the LED Driver.

Figure 6 shows the logic synthesis created for the schematic for the LED Driver digital block. The schematic has 69 instances, being 16 sequential, 3 inverters, 2 buffers, and 48 logics. The results of the logical synthesis for the estimated power consumption and the estimated area of the LED driver are shown in Table 1.

TABLE I: Power consumption and area taken from the logical synthesis.

Parameters	Value
Estimated Power Consumption	73.211 nW
Estimated Area	291 μm^2

As we can check from Table 1, the estimated power consumption is low for the digital block. The estimated power consumption of 73.211 nW is a reasonable value since the state machine only counts with six states. Figure 7 shows the results of the physical synthesis of the LED Driver digital block.

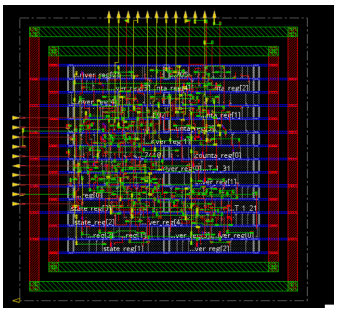

Fig. 7: Physical synthesis for the LED Driver.

From Figure 7, it is possible to check the layout result from the logical synthesis. It counts with the inputs and outputs for the LED Driver digital block.

For the IDAC-PGA digital block, the testbench are shown in Figure 8, Figure 9 and 10 using a clock of 5MHz

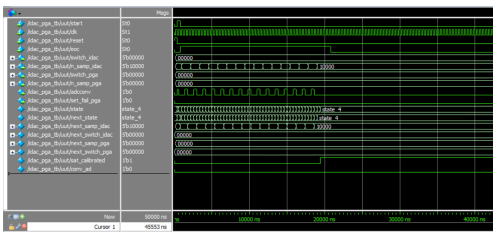

Fig. 8: Testbench for the IDAC-PGA when the ADC is not saturated.

Figure 8 shows the testbench for the IDAC-PGA digital block when the ADC is not saturated. When this occurs, we can check that state 4 is reached and the state machine for the PGA is not activated. For this testbench, the switches were closed and opened until they reached the value of sixteen.

Figure 9 shows the testbench when the IDAC fails to calibrate activating the PGA state machine and shows that if the ADC doesn't saturate for the PGA, no switch will be closed and the state machine will go to state 9.

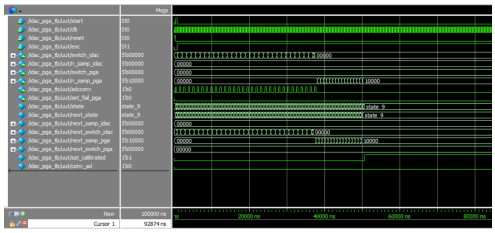

Fig. 9: Testbench for the IDAC-PGA when the ADC is not saturated.

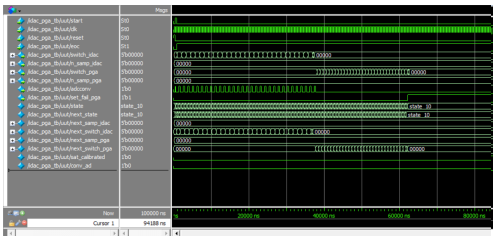

Fig. 10: Testbench for the IDAC-PGA when the ADC is not saturated.

Figure 10 shows the testbench when both IDAC and PGA fail to calibrate the system. If the ADC remains saturated after all combinations of switches for the IDAC and the PGA, it will raise a failure flag and reach state 10, which is the state of calibration failure. Figure 11 shows the logical synthesis for the IDAC-PGA digital block.

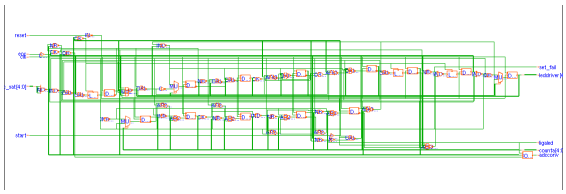

Fig. 11: Logical synthesis for the IDAC-PGA digital block.

Logical synthesis for the IDAC-PGA shows the schematic created for the digital block, where 39 instances were created, namely 28 sequential, 12 inverters, 2 buffers, and 97 logics. From the logical synthesis, we can check the estimated power consumption and estimated area in Table 2.

TABLE II: Power consumption and area taken from the logical synthesis.

Parameters	Value
Estimated Power Consumption	122.32 nW
Estimated Area	559.840 μm^2

From Table 2, we can check the value of 122.32 nW for the IDAC-PGA digital block. The value of the estimated power consumption is higher than that of the LED Driver, since it has more instances. The value obtained is reasonable since it is a machine state that counts with eleven states.

Finally, the physical synthesis is shown in Figure 12.

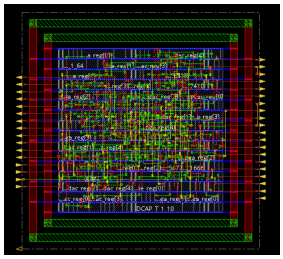

Fig. 12: Logic synthesis for the IDAC-PGA digital block.

Figure 12 shows the result of the physical synthesis in which we can check the layout of the IDAC-PGA digital block. As we can see, the output pins and input pins were created.

TABLE III: Total Power consumption and total area for the Digital Circuit.

Parameters	Value
Total Estimated Power Consumption	195.531 nW
Total Estimated Area	850.84 μm^2

Table 3 shows the total values obtained for the LED Driver and the IDAC-PGA digital blocks.

V. Conclusion

This paper presented the digital control for three blocks that compose the PPGE analog front-end: LED Driver, IDAC, and PGA. Using the theory of state machines, it was possible to create the digital blocks that control the switches for compensating offset and saturation of the signal. From the logical and physical synthesis, it is possible to obtain the schematic for the circuit and the total estimated power consumption of 195.531 nW and estimated layout area of 850.84 μm^2.

References

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